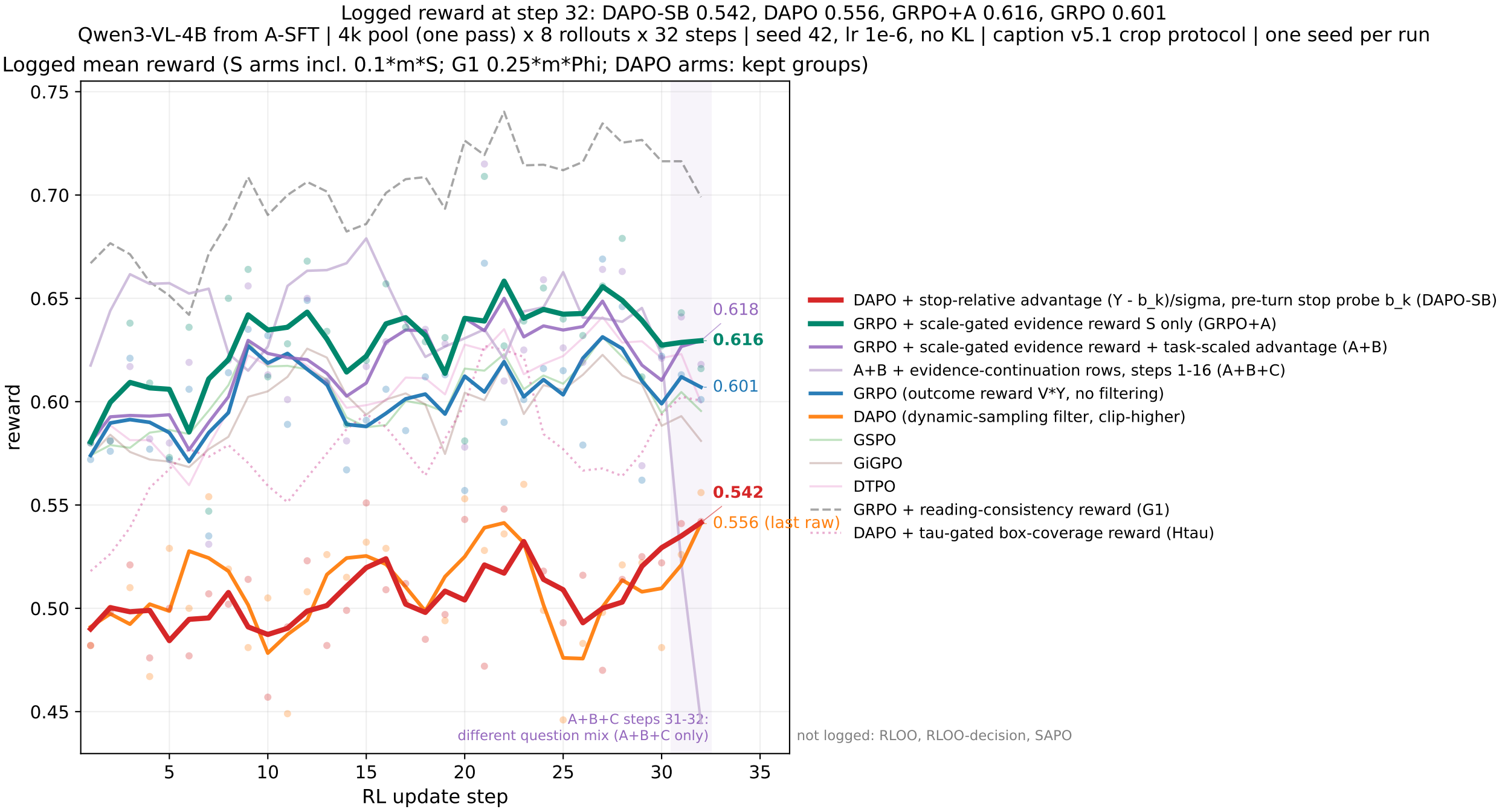
Logged reward at step 32: DAPO-SB 0.542, DAPO 0.556, GRPO+A 0.616, GRPO 0.601
Qwen3-VL-4B from A-SFT | 4k pool (one pass) x 8 rollouts x 32 steps | seed 42, lr 1e-6, no KL | caption v5.1 crop protocol | one seed per run
Logged mean reward (S arms incl. 0.1*m*S; G1 0.25*m*Phi; DAPO arms: kept groups)
reward
RL update step
0.75
0.70
0.65
0.60
0.55
0.50
0.45
5
10
15
20
25
30
35
0.618
0.616
0.601
0.542
0.556 (last raw)
A+B+C steps 31-32:
different question mix (A+B+C only)
DAPO + stop-relative advantage (Y - b_k)/sigma, pre-turn stop probe b_k (DAPO-SB)
GRPO + scale-gated evidence reward S only (GRPO+A)
GRPO + scale-gated evidence reward + task-scaled advantage (A+B)
A+B + evidence-continuation rows, steps 1-16 (A+B+C)
GRPO (outcome reward V*Y, no filtering)
DAPO (dynamic-sampling filter, clip-higher)
GSPO
GiGPO
DTPO
GRPO + reading-consistency reward (G1)
DAPO + tau-gated box-coverage reward (Htau)
not logged: RLOO, RLOO-decision, SAPO